

IN-LIFE OPERATIONS

DEBIAN OPERATING SYSTEM INSTALLATION ON ALLETRA 4120 SERVER

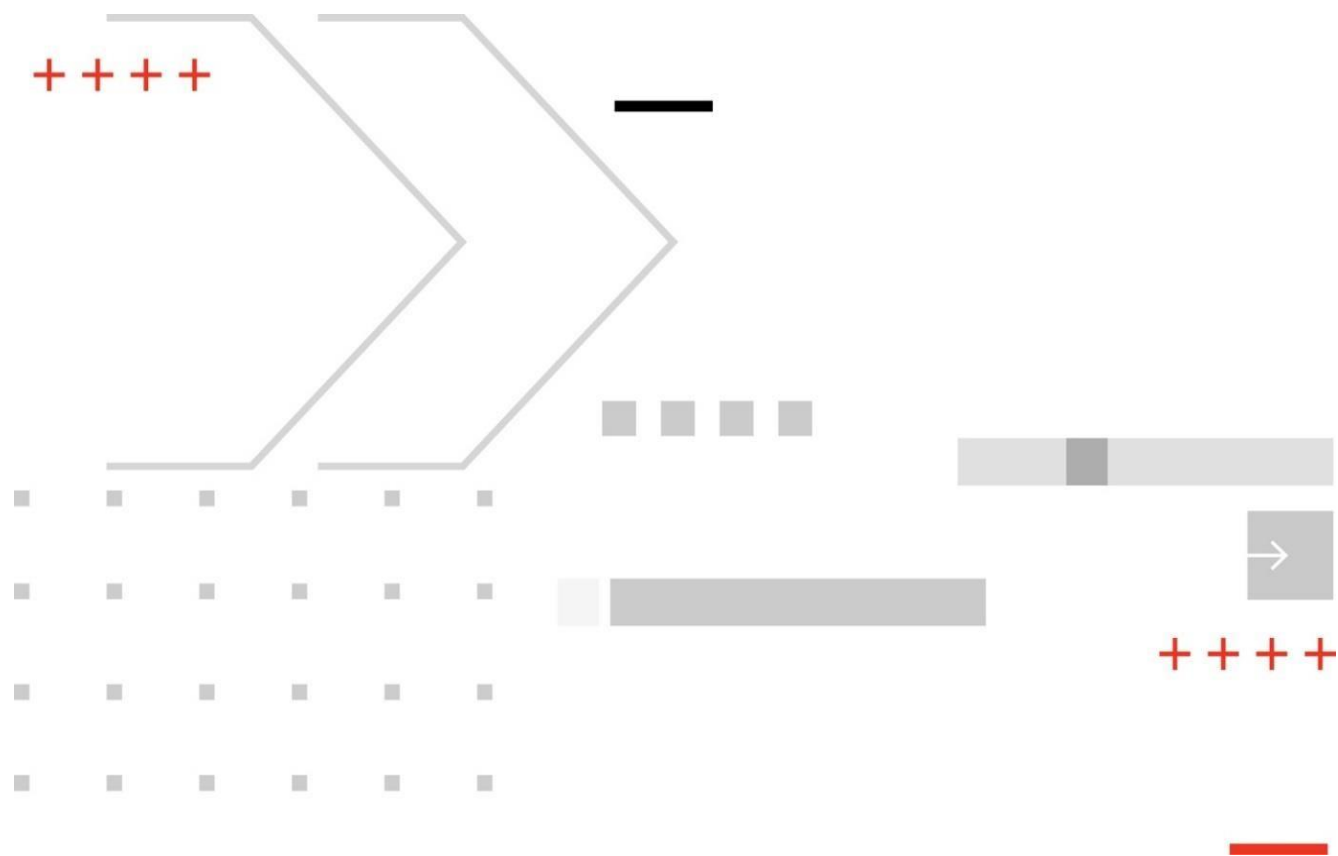
SYNOPSIS: This document describes the procedure for installing the OS on the ALLETRA 4120 server model

REFERENCE: 100346 – Installation and service configuration manual

VERSION: V1.3

DATE: September 2026

STATUS: In Progress



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1. INTRODUCTION

1.1 PURPOSE

Within the framework of infrastructure operations of the ICT CALA L1 team, the Debian 11 Operating System has been installed on the HPE Alletra 4120 server. The purpose of this document is to record in detail the technical procedure followed, from hardware preparation (RAID configuration and arrays), OS deployment through the iLO console, to the initial server configuration.

1.2 SCOPE

The scope of this document is on service management for the MS-Claro Colombia team.

1.3 INTENDED AUDIENCE

This document is intended for use in services intended for MS Claro Colombia by the Product Support team (L1), Managed services team and L2 Support team.

1.4 ESSENTIAL SERVER CONCEPTS

- **Base server:** HPE Alletra 4120 acts as a central computing platform for corporate services (For example: databases, virtualization, critical applications).
- **Operating system:** Debian 11 is selected as the Operating System for its stability, long-term support, active community, and compatibility in production environments.
- **Remote access:** through the server's iLO console, which allows remote hardware control and monitoring, even without an installed Operating System.
- **Storage:** prior configuration of two RAID arrays, each consisting of eight physical disks, managed from the HPE Smart Array Controller.

Taking the above into account, it is established to guarantee the installation of a stable, secure operating system ready for the implementation of services or virtualization environments.

2. Hardware and Software Requirements

Hardware Requirements (Alletra 4120 Server)

Servidor: HPE Alletra 4120, con controladora de almacenamiento, discos configurados (RAID), memoria RAM suficiente, procesador.

Storage: disk space according to inventory availability (multiple disks, boot from RAID controller). In this case RAID arrays:

- RAID 6 + 0 (8 disks – Array A)
- RAID 5 (8 disks – Array B)

Red: compatible network interface, preferably with PXE boot support, static IP address.

Remote access: HPE iLO 5 o superior (montaje de ISO y consola virtual)

Otros: redundant power supply, UPS connection, system ventilation.

Software Requirements

Operating system: Debian 11.

Installation media: ISO image mounted from the iLO interface (no physical media required)

Additional packages: openssh-server, sudo, htop, vim, fial2ban, ufw

Firmware: iLO BIOS for Alletra 4120 (Latest available version)

Additional prerequisites

- Disk partitioning planning (e.g., /, /var, swap, etc.)
- Static IP address for server, defined hostname and domain
- Administrator credentials and roles for the system (root and sudo user)
- Verification of server BIOS/UEFI compatibility for UEFI mode boot or legacy.
- Verify that the server is in the appropriate boot mode and that the installation media is

accessible

3. Installation and service configuration

Below is the step-by-step process for installing Debian 11 on the Alletra 4120 server:

3.1 Preparations

1. Descargar la ISO de Debian 11 (amd64) desde el sitio oficial de Debian. <https://www.debian.org/distrib/>
2. Access the server management console (iLO) and mount the ISO or USB as boot media; in this specific case, it was done using the ISO.
3. Verify that the server is configured to boot from USB or installation drive; if applicable, enable booting in UEFI/Legacy mode according to policy.
4. Verify that the storage (RAID or disks) is ready, and that the required mode is enabled in BIOS/firmware.

3.2 Prior RAID Configuration (Mandatory before installing Debian)

1. Access the server BIOS/UEFI by pressing F9 during boot.

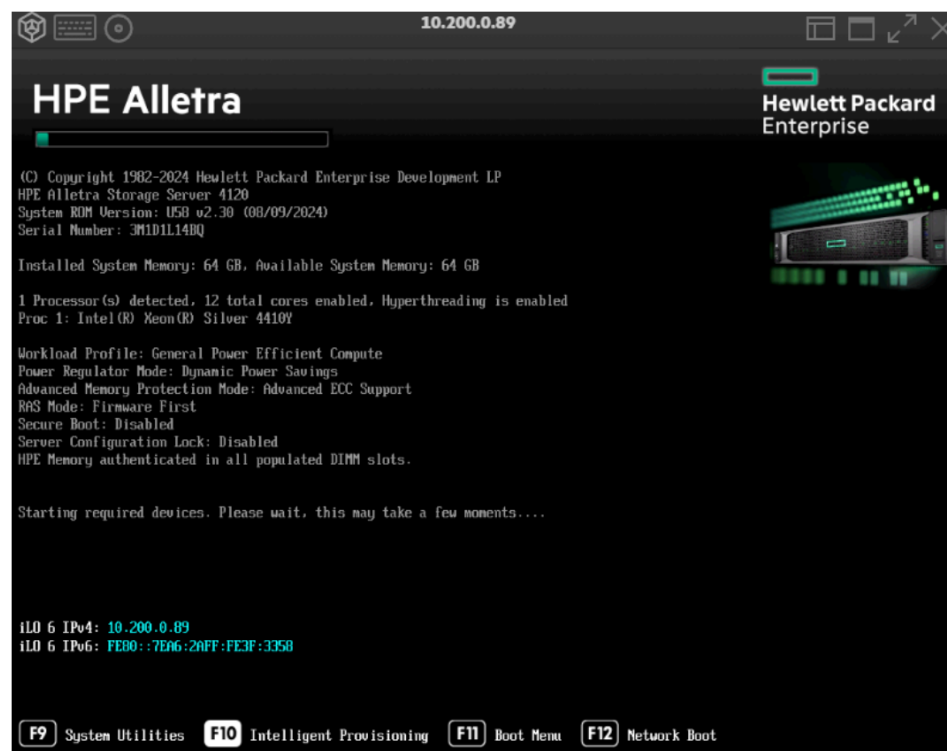


Figura 1: BIOS HPE Alletra 4120

2. Access the Smart Storage Administrator (SSA) or Array Configuration Utility (ACU).

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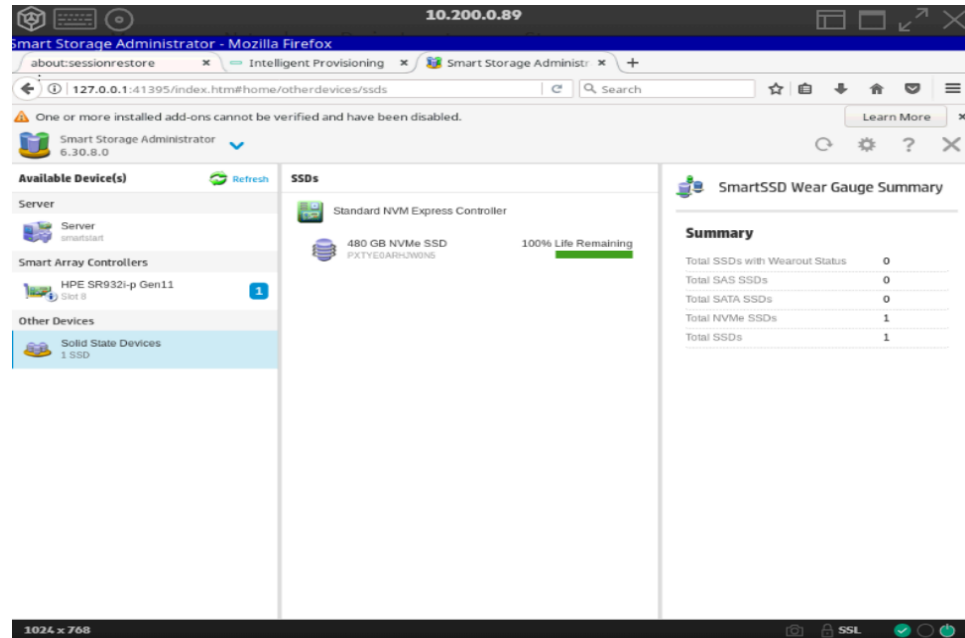


Figure 2: Smart Storage, HPE Alletra 4120

3. Create two RAID arrays: for this specific case, the recommendations from the guides provided by Veeam Backup for configuring an immutable repository were followed, which suggest implementing RAID 60.

RAID 60 is a nested RAID type that combines two configuration levels:

- Level 1: RAID 6, data is grouped into subsets (minimum of 4 disks) and each uses RAID 6 to provide double parity, allowing each RAID 6 subset to continue operating if two disks fail.
- Level 2: RAID 0, these multiple RAID 6 groups are grouped together in a RAID 0 configuration ("striping"). This means that data is divided and distributed among the different RAID 6 groups, improving overall performance.

Configuration Benefits

- Ideal for critical applications: combination of performance and reliability makes it suitable for applications that require low latency and high data security availability
- High performance, through RAID 0 striping distributes data among groups, resulting in stable read and write speeds
- High redundancy and data protection: multiple disk failures are supported, as each RAID 6 subgroup can handle two disk failures.

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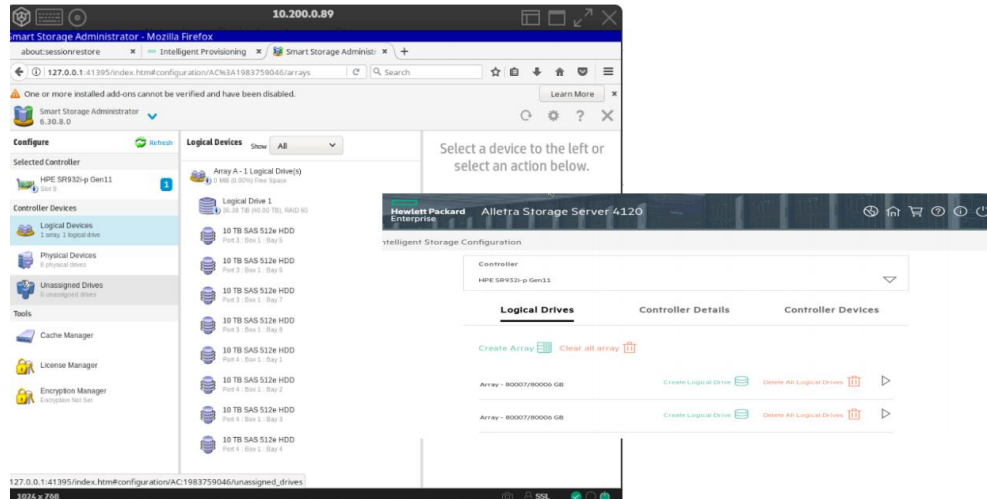


Figure 3. Array A Configuration, HPE Alletra 4120

4. Save the configuration and restart the server.
5. Verify from the BIOS or SSA that the logical volumes are in Healthy status
6. Exit the utility and proceed with normal boot.

3.3 Installation Media Preparation (iLO)

1. Access the server's iLO interface using its management IP;
2. Start Remote Console
3. In the Virtual Media > Image File CD/DVD menu, mount the previously downloaded Debian 11 amd64 ISO image.
4. In the Power > Reset menu, select Power Cycle Server to restart from the mounted ISO
5. Verify that the boot is performed from the virtual device.

3.4 Debian 11 Operating System Installation

1. Once the server is restarted, boot from the installation media
2. In the Debian installer boot menu, select “Install” (This will start the text installer) or “Graphical install” (graphical installer), according to preference.

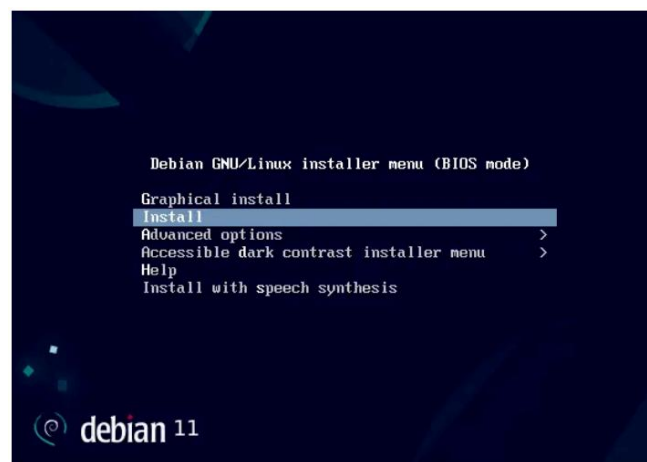
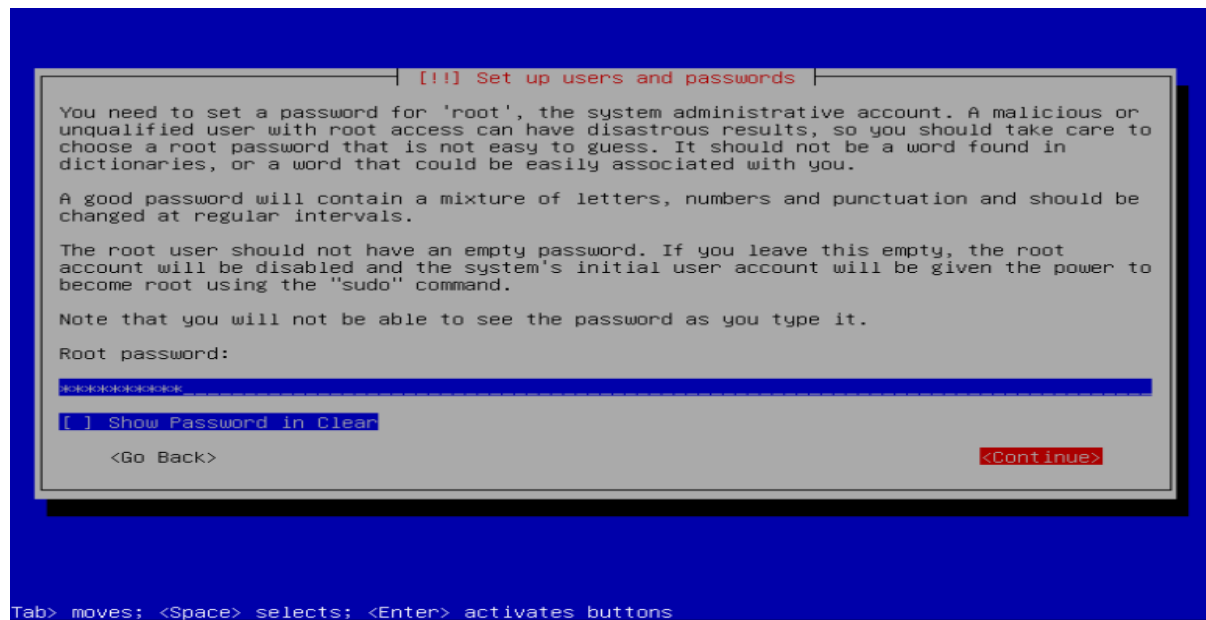


Figure 4. Debian installation interface

3. Then, choose the location and select the keyboard layout. The following screens will vary according to your preferences. Simply select the appropriate country and keyboard layout for you, as they define the language your Debian system will use in the Shell and the keyboard layout used. In this case, the keyboard layout used is – English language, location and keyboard configuration (Latin America).
4. Configure the network: enter hostname (Alletra), domain (@csg.com), IP (10.10.1.189), gateway (10.10.1.254), DNS (DNS_1: 172.16.255.200 and DNS_2: 172.16.255.183) and network mask (255.255.255.0). The installer verifies the installation CD and its hardware and configures the network with DHCP if there is a DHCP server on the network
5. Then, a password is assigned to the root user, it is confirmed, and click continue:



6. Create an administrative user account (alletra) with a secure password.
7. In the disk partitioning stage, select the option “Use entire disk and set up LVM”

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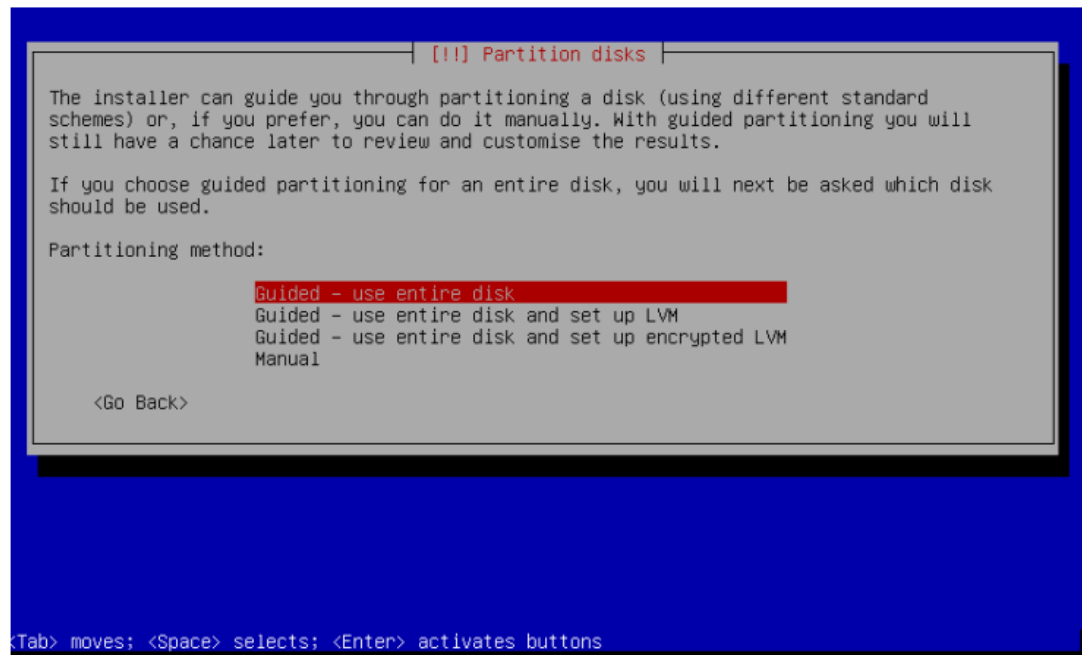


Figure 6: Disk partitioning interface in Debian

8. Select the volume corresponding to RAID 6 + 0 (Array A) and select finish



Figure 7: Disk partitioning interface in Debian

9. Next, configure **apt**. Since we use the Debian network installation CD, which contains only a minimal set of packages, we must use a network server. Select the country where the network server you want to use is located (normally, this is the country where your server is located). Then select the mirror you want to use (**deb.debian.org**).

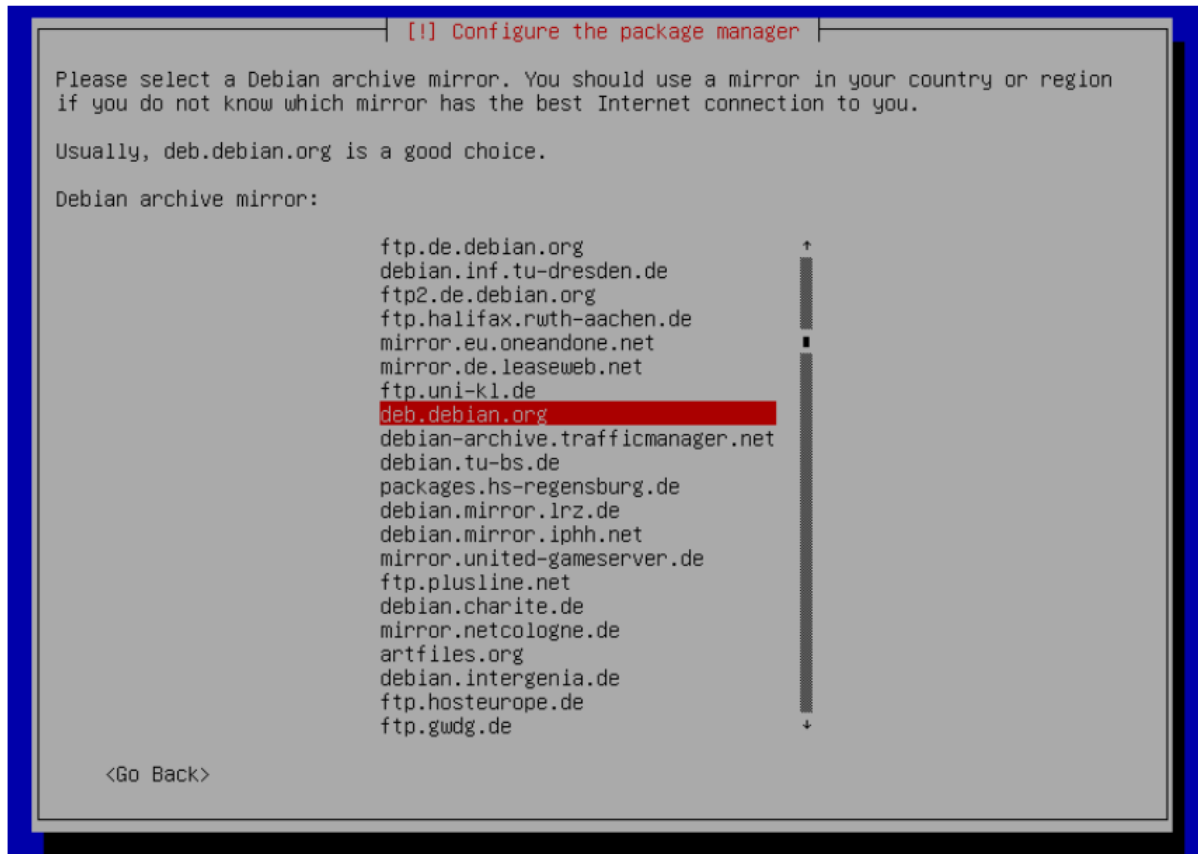


Figure 8. Debian mirror selection

10. Unless you use an HTTP proxy, leave the following field blank and press continue, after which Apt will update the package database.

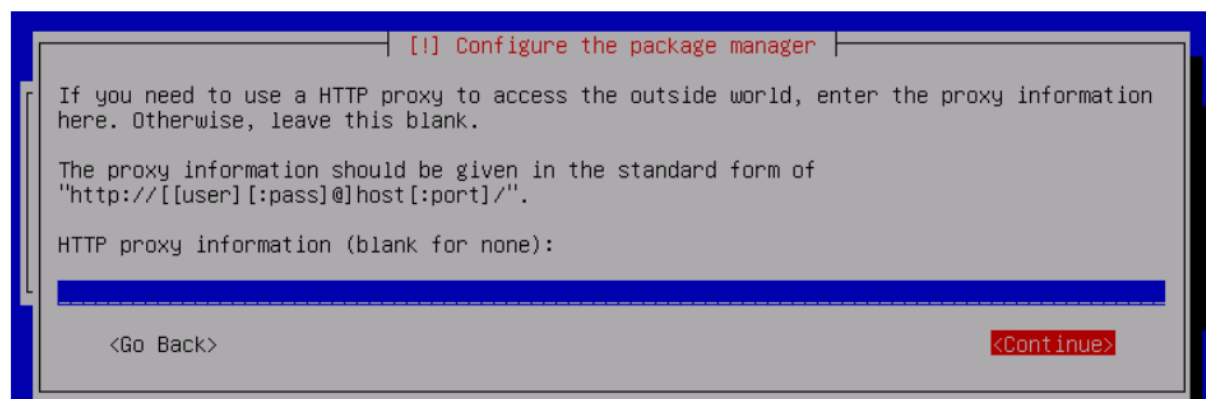


Figure 9. Proxy Configuration

11. Install only the minimum components:

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- Standard System Utilities
- SSH Server

We select standard system utilities and SSH server (to be able to connect immediately to the system with an SSH client like Putty once the installation is finished) and press Continue.

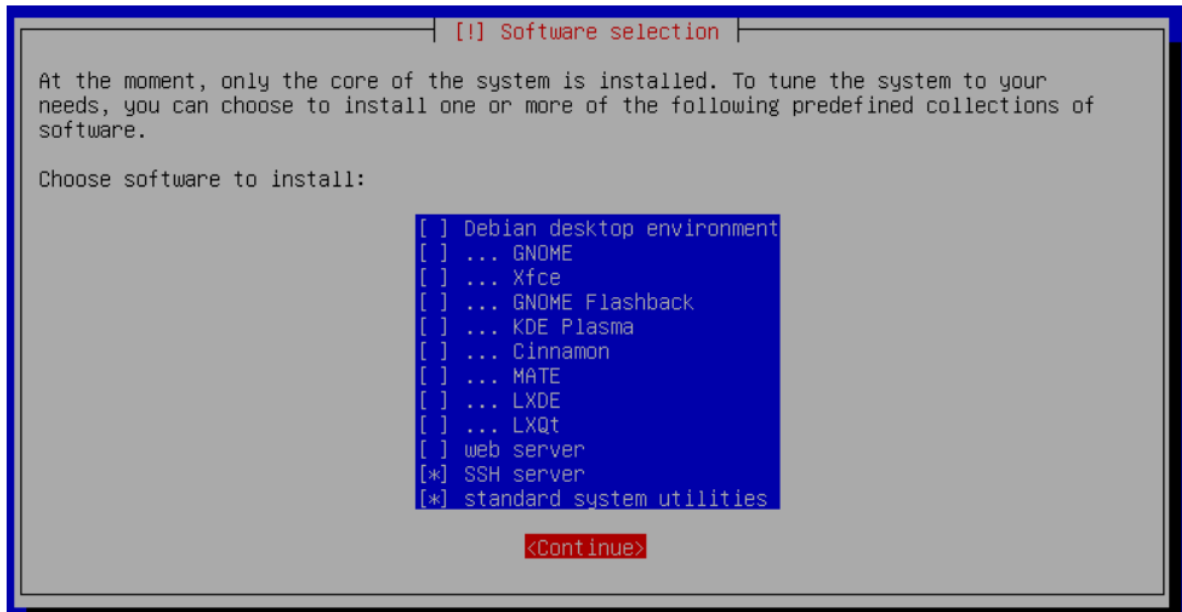


Figure 10. APT Update

12. The necessary packages are downloaded and installed on the system

13. Install the GRUB bootloader on the operating system volume (/dev/sda), to do this select "Yes":

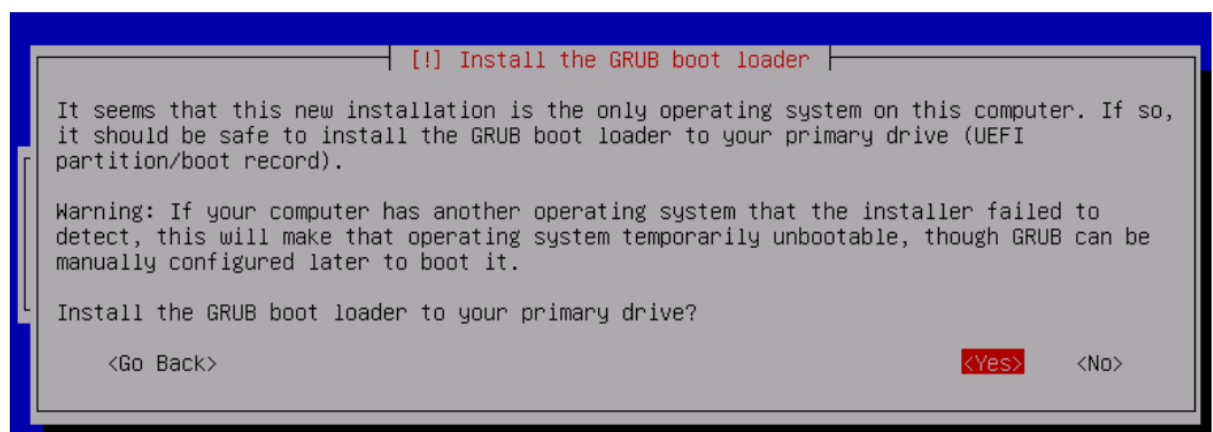


Figure 11. Install GRUB

14. Press **Enter** for the installation of **GRUB**, the installation will be completed and the system will restart.

15. On the first boot of the Debian server: you will see the GRUB boot loader boot screen, press **Enter**

or wait a few seconds and the boot process will continue automatically.

3.4 Post-installation Configuration

The following steps are followed to establish the post-OS configuration on the ALLETRA 4120 server.

- 1. Log in as root or the created user (Alletra).
- 2. Download sudo, with the following command:

```
$ sudo apt-get install sudo
```

- 3. Install the SSH server

```
$ apt -y install ssh openssh-server
```

From now on, you can use an SSH client like Putty and connect from the workstation to the Debian server.

The Static IP is configured by editing the /etc/network/interfaces route, in case it is not configured, as follows:

Table 1. IPv4 address assignment

IPV4	10.10.1.189
Gateway	10.10.1.254
DNS:	DNS1: 172.16.255.200 DNS2: 172.16.255.183
Network mask	255.255.255.0

```
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

# The primary network interface
auto ens33
iface ens33 inet static
    address 192.168.0.100
    netmask 255.255.255.0
    network 192.168.0.0
    broadcast 192.168.0.255
    gateway 192.168.0.1

# This is an autoconfigured IPv6 interface
iface ens33 inet6 auto
```

Figure 12. /etc/network/interfaces

3.5 Boot Media Configuration with Debian

Once the Debian operating system installation is complete, it is necessary to configure it as the server's primary boot media. To do this, the following steps must be followed:

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1. Restart the server and enter the BIOS by pressing the F9 key to access the system utilities.



Figure 13. BIOS HPE Alletra 4120

1. In the main menu, select **System Utilities** y luego acceder a **System Configuration**

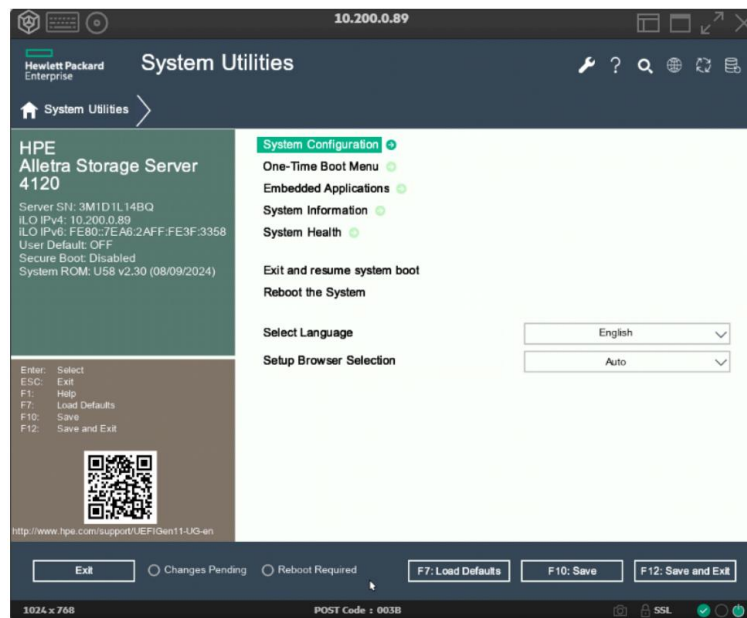


Figure 14. System Utilities Menú

2. Access the BIOS/Platform Configuration (RBSU) option
3. Select Boot Options, where all the settings related to the boot server are located.
4. Enter UEFI Boot Settings
5. Enter UEFI Boot Order, section in which the order of the server boot methods is defined
6. In this section, all available boot media are listed. As shown in the figure, the recently installed system (Debian) should appear. Select Debian and move it to the first place in the boot order to set it as the main medium.

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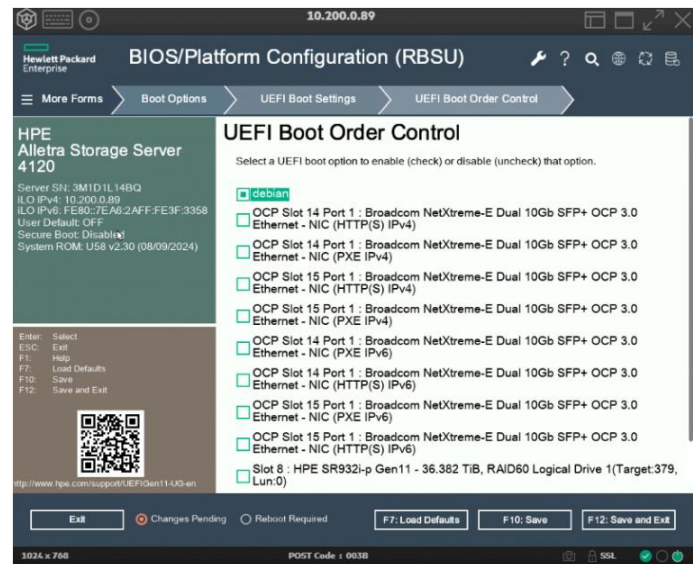


Figure 15. UEFI Boot Order Menú

7. Finally, press F10 to save the changes, verify that it has been saved, and exit the configuration.

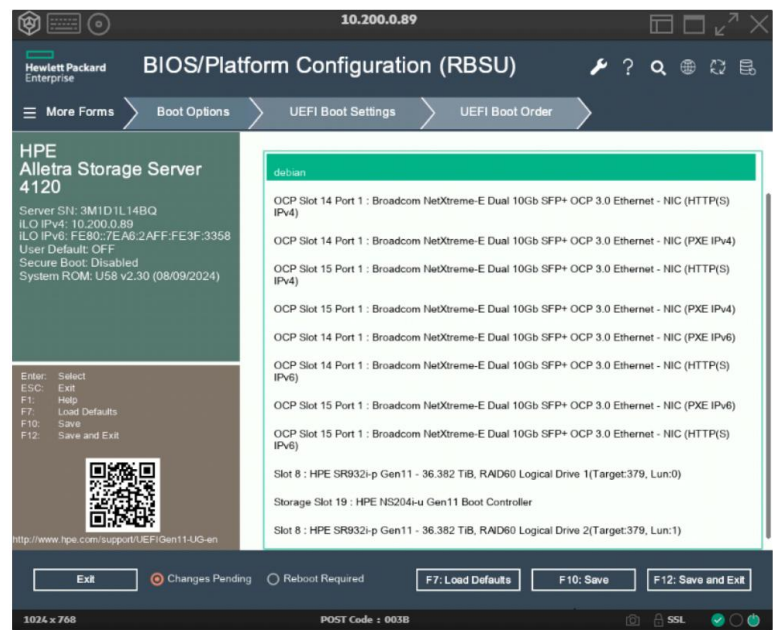


Figure 16. BIOS/Platform Configuration Menu

4. Risk factors in procedure

Problema	Cause	Impacto	Applied Solution
Initially Ubuntu was installed because the HPE manual recommended it, but Debian 11 was ultimately chosen for its stability and small size.	Inconsistency between manufacturer's recommendation and installation restrictions in corporate infrastructure.	Delay in defining the operating system to use, adjustments to the documented procedure.	Debian 11 is definitively implemented by joint decision of the IT infrastructure team and PSAI taking into account data accessibility policies.

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During the Debian installation process, the system did not allow downloading certain components because it was not connected to the Internet.	The network card (NIC)/link to the server switch had no Internet access at the time of installation.	Installation stopped due to lack of connectivity, several days of waiting while the Switch was diagnosed and changed.	The defective Switch was replaced with a functional one, connectivity was verified, then the installation continued normally.
When configuring the Debian boot system, the server did not boot correctly: it went directly to the BIOS/UEFI instead of booting the operating system.	The boot configuration in UEFI/BIOS mode did not point to the correct disk or RAID logical volume.	Inability to start the operating system, delay in start-up.	The boot configuration was adjusted in BIOS/UEFI (UEFI mode, boot priority). These actions were supported by Debian boot troubleshooting guides.

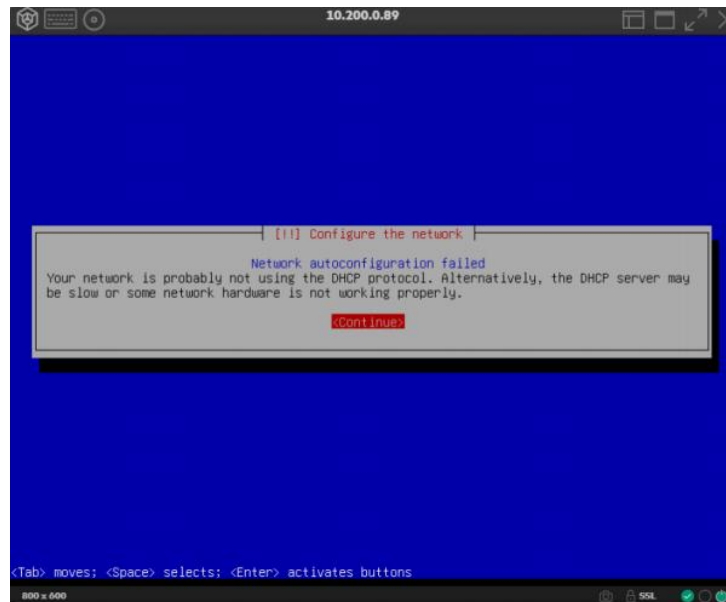


Figure 17. Error No 2, Conectivity problems

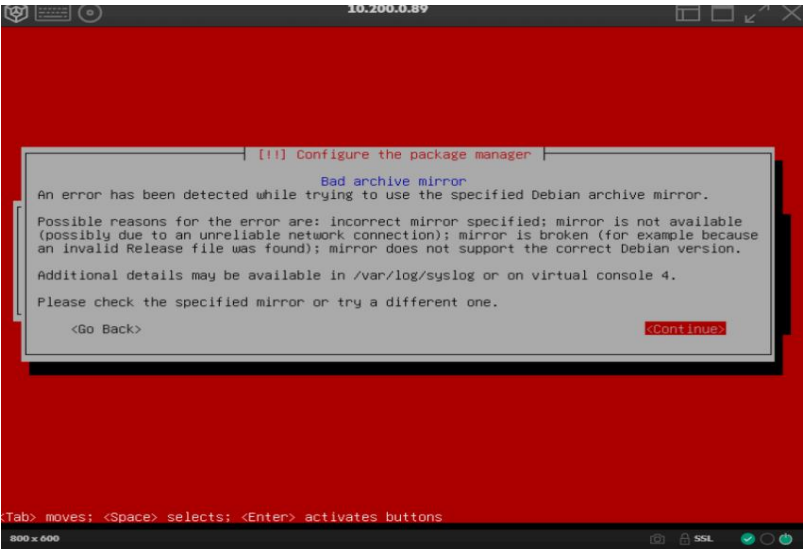


Figure 18. Error No. 2, Network mirroring error due to connectivity issues

5. Test plan

Prueba	Description	Expected result	Obtained Result
RAID Verification	Confirmation of “Healthy” status in iLO or SSA	Approved	Approved
Boot from ISO	El servidor inicia correctamente the installation	Approved	Carrying out additional adjustments - Approved
Base installation	Debian 11 installed without errors	Approved	Hardware device change (Switch) - Approved
post-installation boot	Sistema inicia sin medio de installation	Approved	Carrying out additional adjustments - Approved
Network Interface Configuration	Static IP funcional, acceso SSH activo	Approved	Approved
Firewall	Solo puerto 22 abierto	Approved	Approved
Integrity of volumes	Lsblk y df -h show the estado de los discos actuales	Approved	Approved

6. Version control

Versión	Date	Author	Descripción
1.0	03/09/2025	Pablo César G.	Firmware update
1.1	10/09/2025	Pablo César G.	Instalación de 8 discos duros
1.2	17/09/2025	Pablo César G.	Switch position change
1.3	24/09/2025	Pablo César G.	Debian OS installation